INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

Distributed Systems

Introduction

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Recap

- Distributed Systems - Issues / Limitations
- Two army problem & TCP connection establishment
- Distributed Search
- Transparency in distributed systems

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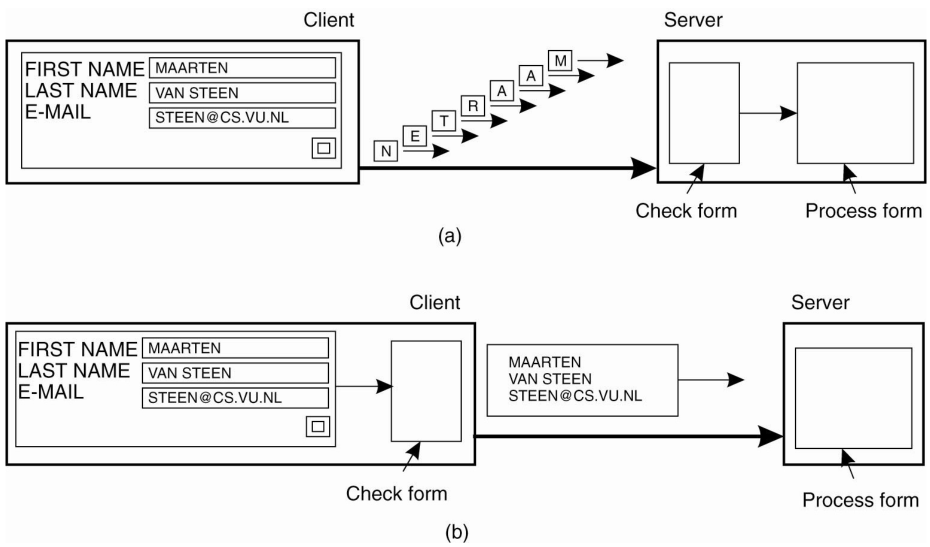
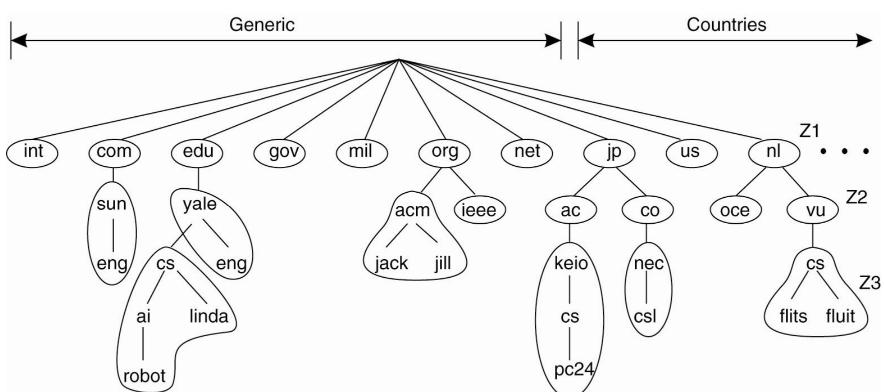
Scaling Techniques [5]



Scaling Techniques [5]



DNS name space

DNS Name Resolution (DNS Lookup)

URL
Domain
TLD

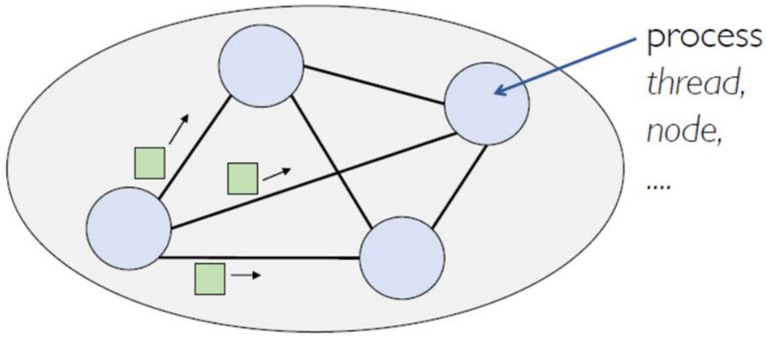
Pitfalls when Developing Distributed Systems

- The network is reliable
- The network is homogeneous
- The topology does not change
- Latency is zero
- Bandwidth is infinite
- Transport cost is zero
- There is one administrator
- The network is secure

Inherent limitations of distributed systems

- Lack of common memory
- Absence of a global (system wide) clock
- Unpredictable communication delays

Distributed system

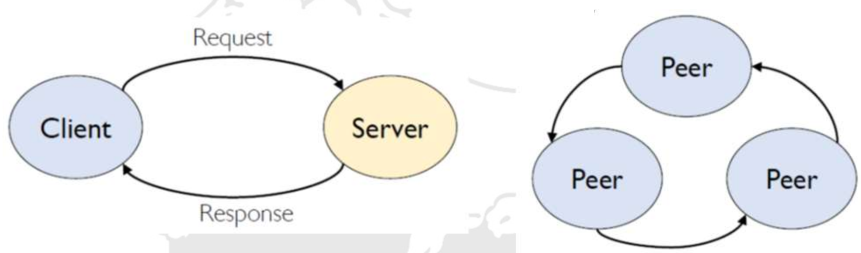


- A set of autonomous processes communicating among themselves to perform a task
  - to achieve a common goal
  - appearing as a single coherent system

Relationship between processes

- Two main categories:

1. Client-server
2. Peer-to-peer



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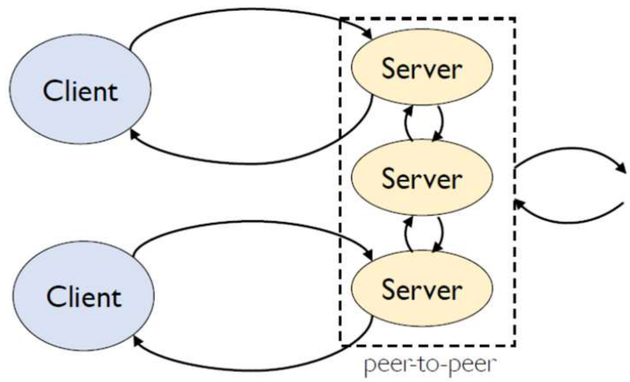
Relationship between processes



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Relationship between processes

- Most network applications can be divided into two pieces: a **client and a server**.
  - Web browser communicating with a Web Server
  - FTP client fetching a file from a FTP server
  - Telnet client and Telnet server
- Client and Servers may be on the same LAN or on a WAN
- Communication between client and server involves networking protocols (OSI, TCP / IP)
- A program / node that can act both as a client and a server is based on **peer-to-peer** communication

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P2P

Issues:

- **Connectivity**: how to find and connect other P2P nodes
- **Instability**: nodes may always be joining and leaving the network
- **Message routing**: how messages should be routed to get from one node to another
- **Searching**: how to find desired information from the nodes connected to the network
- **Security**: nodes must be able to trust other nodes

Client-server vs P2P

Peer-to-peer:

- Relatively inexpensive
- Fairly simple to set up and manage
- Overburdens user PC by having them play the role of server to other users
- Largely unsecured
- Unable to provide system-wide services

Client-server:

- Higher initial investment
- Greater level of expertise required to configure and manage
- Allows user workstations to function as unburdened clients
- Configurable for maximum security
- Can provide sophisticated system-wide services

Examples of distributed system

- Client-Server or P2P?
  - Web Search
  - Stock Exchange
  - Massively Multiplayer online games / music / films
  - Travel Reservations
  - Skype
  - WhatsApp messaging
  - YouTube
  - Bitcoin

Some Fundamental Problems

- Ordering events in the absence of a global clock
- Capturing the global state
- Mutual exclusion
- Leader election
- Clock synchronization
- Termination detection
- Agreement protocols

Absence of Global Time

- Scheduling processes (temporal order)
- Collecting up-to-date information on the state of the system

Absence of shared memory

- A process in a distributed system can obtain a coherent but partial view of the system or a complete but incoherent view of the system